
Pole Beans

Pole beans offer faster picking, much higher yields and longer harvest periods than bush beans and can be profitably harvested for 6 to 10 weeks, if grown the right way. They can be cultivated both in the field and in a greenhouse and will grow on a treillis.

All and all, they don't generate as much revenue as other greenhouse crops like cucumbers, tomatoes or bell peppers. However, they are very popular at the market and they'll attract a lot of clients that will also buy other products from your stall. Since they require constant temperature, if you have extra space in your greenhouse they are worth it. If not, focus on planting more profitable greenhouse crops and consider growing bush beans in the field instead.

Note that many pole beans varieties are developed for greenhouse production, and so they are more sensitive to temperature fluctuations. Night temperatures below 50°F (10°C) inhibit growth and flowering, same for temperatures above 86°F (30°C).

Family

- Fabaceae

Cultivars

- Green beans: Emerite, Seychelles, Cobra
- Yellow beans: Monte Gusto

Intensive Spacing

- 1 row per bed every 12 in (30 cm) on the row - 3 seeds per cell.

Starting Seeds

- 2 weeks before transplant, sow 3 seeds per cell in 72 plug flats. Seeds should be about ½ in (1.3 cm) deep in the cell.
- Place trays on heating mats and maintain soil temperature at 86°F (30°C) until emergence. At that time, the potting-mix must be kept moist at all times.
- After emergence, the ambient temperature in the nursery can be kept around 75°F (24°C) during the day and 65°F (18°C) at night.
- If you are not transplanting the bean seedlings in a controlled temperature greenhouse, don't forget to acclimatize the plants before transplanting. We do this by exposing them to outside temperatures and reducing the water intake 4 days before the transplant date.

Crop Planning

We plan to grow greenhouse pole beans two times in the growing season. We plan for a first generation in early spring to be the first at the market with beans. Then, we plan a second generation in the fall, when it gets too cold in the field for the beans. In between these two generations, we plan for 1 or 2 generations of bush beans grown directly in the field.

If you don't have access to a greenhouse, it is also possible to grow pole beans in the field as long as temperatures aren't under 65°F (18°C) on average.

When deciding to grow them continuously throughout the season, a new seedling can be started every 6 weeks.

Fertilization

- Unlike bush beans, pole beans produce over a longer period, and will be kept in the ground for a very long time (+/- 100 days). We front-load the

total amount of fertility requirements at the initial bed prep stage so it's not necessary to fertilize them afterwards.

- Apply lots of compost, +/- 2 in (5 cm) all across the bed along with a dose of nitrogen-rich organic fertilizers like pelleted chicken manure (5-3-2) and alfalfa meal (2-0-2). Also, beans will appreciate extra potassium which helps with overall yields. We use organic Sul-Po-Mag (0-0-22).

Irrigation

Irrigation Type

- Drip tapes only (4 per bed)
- No sprinkler to keep foliage and pods dry and to prevent fungal diseases.

Irrigation Management

- Root depth: Shallow (6 to 24 in or 15 to 61 cm)
- General: Rapid growth and ongoing flower production requires frequent and consistent watering.
- It is critical to water consistently at the flowering and fruit development stages.

If you are growing pole beans in a greenhouse, set the irrigation on an automated watering clock. Program three different irrigation cycles:

- Sunny (6 irrigation cycles per day)
- Partially sunny (4 irrigation cycles per day)
- Cloudy (2 irrigation cycles per day)

To manage this, set different start times and cycles on the clock. The irrigation is adjusted each morning according to the outside temperature. It takes only a few seconds per day to adjust the water clock.

Irrigation cycles begin when the plants are active: 3 hours after sunrise if it's a sunny day or 4 hours after sunrise if it's cloudy. Cycles should stop 6 hours before sunset. The length of each irrigation pulse has to be adjusted to ensure that the plants are getting the right amount of water according to their needs (those will vary with different factors like outside temperature and plant size). A good way to confirm your irrigation cycles is to sample soil in the greenhouse at the end of the day and the first thing the next morning. When taking a handful of soil, both should feel about the same.

Implantation

Pole beans are grown on strings or on a treillis (we recommend the Hortonova net). We transplant the plant on soil covered with landscape fabric.

The success of pole beans will also depend on ideal soil temperatures which need to be at least 60°F (16°C). This is particularly important for the first generation implanted in early spring.

Equipment

- Power harrow
- 30 in (75 cm) bed preparation rake
- Landscape fabric, staples, drip tape and fittings
- For the treillis set-up: T-posts every 8 ft (2.5 m), post pounder, bailing twine, treillis, 1 in (or more) top bar, chain link fences T & zip ties
- For the string set-up : bailing twine and tomato clips

Set-up and Transplant

We grow pole beans on a standard set-up of 30 in (75 cm) beds with 12 in (30 cm) pathways in the greenhouse and 18 in (45 cm) pathways in the fields. The crop is always transplanted.

- In the case of a transplant in early spring, plan to measure soil temperature one or two weeks prior to the transplant. If soil temperature is below 60°F (16°C), cover the bed with a clear plastic one or two week prior to the transplant date to increase soil temperature. It might also be necessary to start the greenhouse heating system.
- When ready to transplant, remove large debris from the bed if needed (use a flail mower, a rake or manually remove the large chunks).
- Apply compost and other amendments in the middle of the bed and pass with the power harrow at a depth of 2.5 in (6 cm). Since the bed will be covered with landscape fabric, there is no need to clean it up with another pass of the rake.
- Install 4 lines of drip tape per bed and start the irrigation system. Fix leaks at that point, if any.
- Lay down landscape fabric and secure it with staples each second step (5 ft or 1.5 m). It needs to be really tight. Pole beans are grown on a row with 12 in (30 cm) spacing between each set of 3 plants. Holes should be 2 in (5 cm) in diameters.
- Set support posts (T-posts) 16 in (40 cm) into the ground at a maximum distance of 8 feet (2.5 m) apart.
- Set-up the top bar (railing post) by using chain link fences T.
- Secure the trellis to the first support post and unroll it all the way to end. Attach using cable zip ties. Stretch these to their maximum.
- To strengthen your trellis, drive a grade stake into the ground at a 45-degree angle 3 feet from your first and last posts. Attach a line from the top of the grade stake to the top of your first and last post where

the cable is attached. This increases the rigidity of the trellis once it is filled with beans.

- Using zip ties, deploy the trellis net to its maximum stretch. Secure the bottom of the trellis by passing a string through it and tying it to the first and last posts. The trellis is now ready.
- Transplant beans carefully by making sure to place them in the center of each hole. Handle them with care during the transplant and don't bury their collars.
- If using the string set-up, simply tie one string every 12 in (30 cm) on the top wire of your greenhouse. Attach the strings to the bean plants by using a tomato clip. Avoid tying the plants too tight at this step; doing so may uproot them.
- Water the bed thoroughly the first time, keeping the soil moist for the first week after the transplant.
- Note that at that point, the minimum soil temperature is 60°F (16°C), but 68°F (20°C) would be optimal.

+/- 8 Weeks After Transplant

Once your pole bean plants reach the top wire, cut the head of the plants that grow past the top wire or string. Perform this step 2 or 3 times during the growing season. Doing so helps the plants concentrate their energy on completing their fruiting stage.

Climate Management

- If you grow pole beans in a heated greenhouse, control the temperature to favor optimal yields. Change your set points according to outside weather.

Outside weather	Daytime set-point	Night time set-point
Cloudy	65°F (18°C)	65°F (18°C)
Sunny	73°F (23°C) *open roof and roll-ups when inside temperature reaches 80°F (27°C)	73°F (23°C)

Weeding

- No weeding required.

Plant Protection

- Pole beans can be affected by different fungal diseases (*sclerotinia*, *pythium*, *fusarium*, *anthracnose*, etc.). To decrease the risks of infection, good practices include a crop rotation, choosing resistant cultivars, avoiding humid conditions (plan for dehumidification cycles in your heated greenhouse), transplanting into a well drained soil and monitoring the irrigation closely to avoid overwatering the plants. If these practices fail and you end up with a fungal disease, many biofungicides are certified for organic agriculture and can be very effective.

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- If your soil has a history of fungal disease infections, a preventive application of Rootshield (*Trichoderma harzianum*) at transplant is a good practice.
 - Pole beans grown in a heated greenhouse are prone to thrips and two-spotted spider mites attacks. **See the Beneficial Greenhouse Insects module** for more information.

Pollination

Pole beans have perfect flowers that are self-pollinating. For this reason, introducing pollinators in the greenhouse is not necessary, although it could lead to slight increases in yields.

Harvest

- Beans are harvested by moving up and down the plant profile and moving forward using tree planting bags.
- Beans are harvested twice a week during peak season.
- Harvest bean pods when they are 6-7 in long (15–18 cm) and as wide as a pencil at most. You want to pick them before the seeds become bulbous and more visible in the pod.
- You want to be able to harvest the pods with only one hand. This technique will allow you to use both hands simultaneously. To harvest pole beans, pinch the peduncle between your thumb and index. Wrap gently your remaining fingers around the pod, then snap the bean off the peduncle by pushing your thumb on your index.
- Make sure to gather as many beans as you can in each of your hands before putting them in your harvest bag.
- Pods should be clean and free from any mold or rotting.

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- When harvesting plants that are well supplied with beans, plan an average of 2 hours for one person for a 100 ft row.

Conditioning

- Freshly picked beans should not be washed or they can otherwise become moldy. Beans should be cooled in the cold room in sealed bins.
- If the beans do get damp, open the bins and let them dry in the cold room for a couple of hours (max 4 hours as they may decrease in quality).

Storing

- Beans are stored in airtight bins and kept in the cold room at 39°F (4°C). They should not stay longer than 4 days before being sold.